

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

SKILLS

- Languages** – Python, Go (Golang), SQL, C, C++.
- Backend & APIs** – FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio, Design Patterns, Service-Oriented Architecture (SOA).
- AI & Agents** – LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- Data & Streaming** – Kafka, Flink, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- Cloud & Infra** – GCP, AWS, Docker, Kubernetes, Linux, CI/CD, Bazel, Observability (ELK, New Relic, Datadog, PostHog, LangSmith).
- Core** – Distributed Systems, Microservices, System Design (HLD/LLD), Multithreading, Concurrency, Caching, Sharding, Mentorship.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building AssortSmart so retailers decide **what to buy, how much, and for which stores** via a **Go (Gin)** write/doing path behind a planner copilot (**FastAPI, LangGraph, MCP**), targeting plan turnaround from days toward **under 1 hour** and search from **1 to ≥ 20 configs/plan**.
- Cutting measured clustering-run failures from **8.5% (37/437 on kik)** toward **under 2%** (target) with **14** audited read-only tools and **3** human confirm gates so the agent never writes SQL.
- Replacing a projected **12B-row** store-week design with a **$\sim 25\text{M}$ -row** aggregate so month rollups stay **sub-second** and cell edits stay **$\sim 0.4\text{ ms}$** (measured) without materializing the explosion.
- Adopting per-tenant **ClickHouse (63 tables / 8 layers, insert-only / partition-swapped, agent read-only)** as the agentic planning store after a row-identical POC cut **250M-row** pivots from **189s to 12.3s** ($\sim 15.5\text{x}$, measured).
- Tech** – Go, Gin, Python, FastAPI, LangGraph, MCP, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned design and architecture for Uber's **FRM Risk Scoping platform (FastAPI, MySQL, React)** across **8 screens and 30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70% cut** in manual reconciliation (**~ 2 weeks to $\sim 3\text{--}4$ days**).
- Designed an **11-table SQLAlchemy 2.0 MySQL** schema with handler, service, and repository layers that auto-flags material items across **55 line items and 14 entities**.
- Owned Sheets to **MySQL v2 recon migration** across **18 files**, validating HFM extracts against public **10-Q** filings and fixing a latent column-aliasing bug.
- Led 3 engineers** through design reviews, API contracts, and CI quality gates for releases feeding PwC audit work papers.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Cut menu onboarding from **24h to 2h** and saved **\$600K+ annually** on **30K+ menus/month** by shipping **Selenium** scrapers on GCP with **Kafka** ingest and **Flink** keyed normalize/dedupe and replay.
- Converted unstructured multilingual menus (**PDFs/images**) into Uber Eats schema at **98% fidelity / 100% schema consistency** (offline eval) via a **LangChain RAG** pipeline with **Gemini 2.5 Pro** over a **vector store** of labeled menus, plus supervised fine-tuning.
- Raised successful ingestions to **95%+** by handling anti-bot defenses with IP rotation, dynamic proxy pools, and adaptive backoff.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, vector DB, SFT, GCP, Docker.

ANZ Driver Document Compliance (Uber Mobility)

- Cut manual verification by **20 hours/week** by automating **driver and vehicle document compliance to 99.9%** for **Uber drivers in ANZ** against local authorities.

Software Development Engineer 2 MASTERS INDIA, Noida December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Cut p95 latency from **1.2s to 300ms** for **1,500+ clients** by strangling a PHP (**Laravel**) monolith into **FastAPI microservices** and mentoring **2 engineers**.
- Lifted throughput from **700 to 4,000 requests/min** and processed **1M+ IRP submissions/day (100K+/import)** by building an event-driven path on **Kafka** and **quarter-sharded PostgreSQL**.
- Cut redundant DB reads **30%** and made IRP replay safe by shipping **idempotent sinks (idempotency keys, retries, DLQ)** with Redis cache-aside.
- Cut incident triage **70%** and support tickets **35%** by shipping **ELK / New Relic** on-call alerting plus a client usage dashboard, raising coverage **35% to 82%** at **98%** deploy success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer GEEKSFORGEEKS, Noida August 2021 – November 2022

- Stabilized doubt-support for **10K+ daily queries** and **10x** contest spikes by migrating the backend from PHP to **Django (Python)**.
- Lifted premium subscriptions **15–20%** by shipping voting, pinning, and locking REST APIs on MySQL, MongoDB, Redis, and Elasticsearch.
- Raised course sales **30%** by launching an influencer earnings and analytics dashboard (transactions, coupons, filters).
- Raised operational efficiency **70%** by orchestrating cron pipelines for video processing, reminders, and recording cleanup.

EDUCATION

B.Tech (Information Technology) Institute of Engineering & Technology, Lucknow July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).